Fresh Snapshot Evaluation Before Feedback Collection

10,000 unseen URLs • 5% phishing • frozen models, checkpoints and thresholds

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	97.20%	86.66%	85.10%	85.87%	97.03%	91.34%	1.46%	14.90%
HYB-MS +10% GAN	97.84%	89.92%	88.30%	89.10%	97.45%	93.51%	1.10%	11.70%

Outcome

The +10% GAN model improved every reported metric, including recall, F1 and ROC-AUC, while lowering FPR. The registered augmentation rule passed.

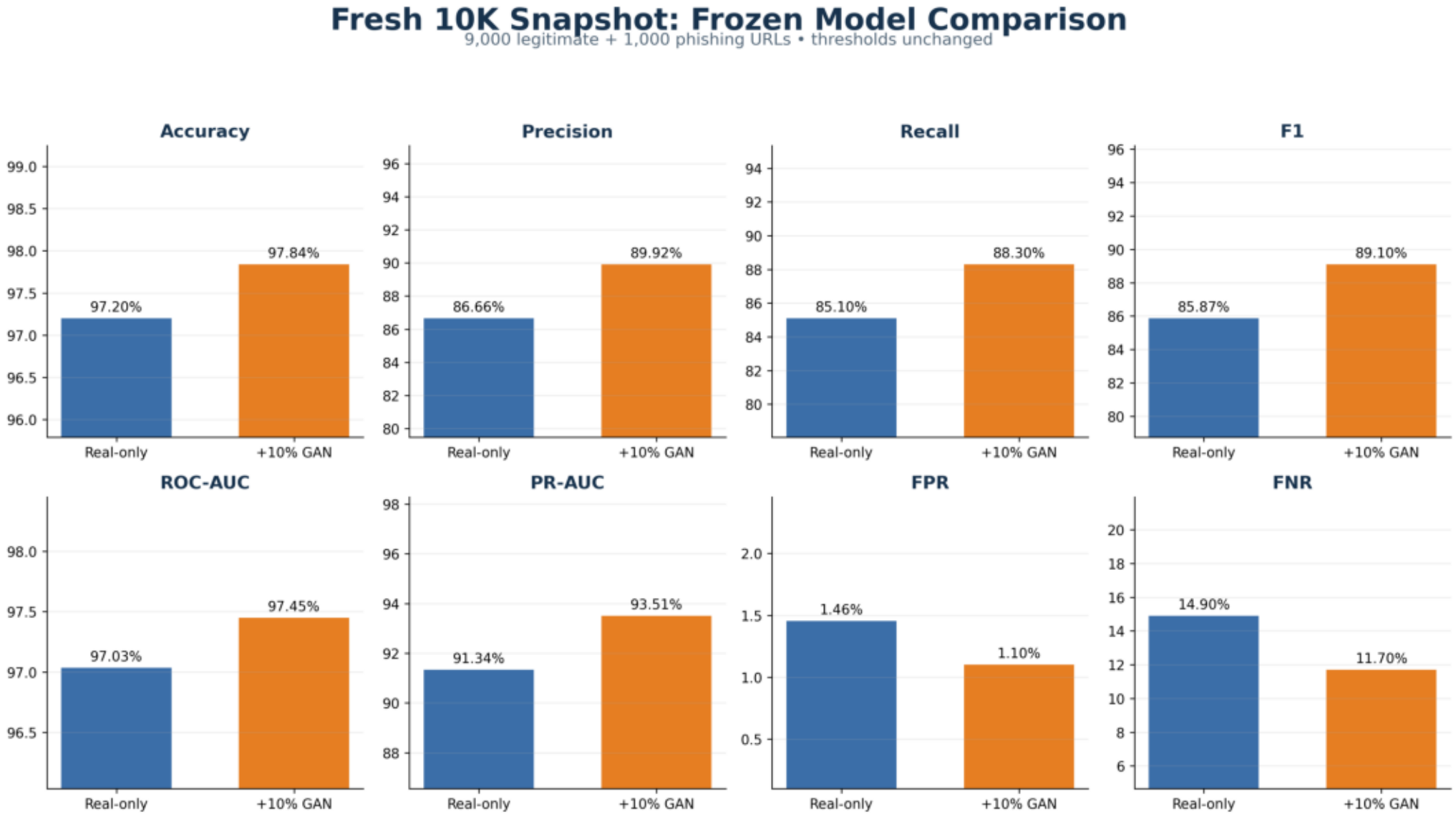
GAN changes: F1 +3.23% • Recall +3.20% • FPR -0.36% • FNR -3.20% • ROC-AUC +0.41%

Deployment metric gate: both models exceeded 80% recall and stayed below the 2% FPR ceiling.

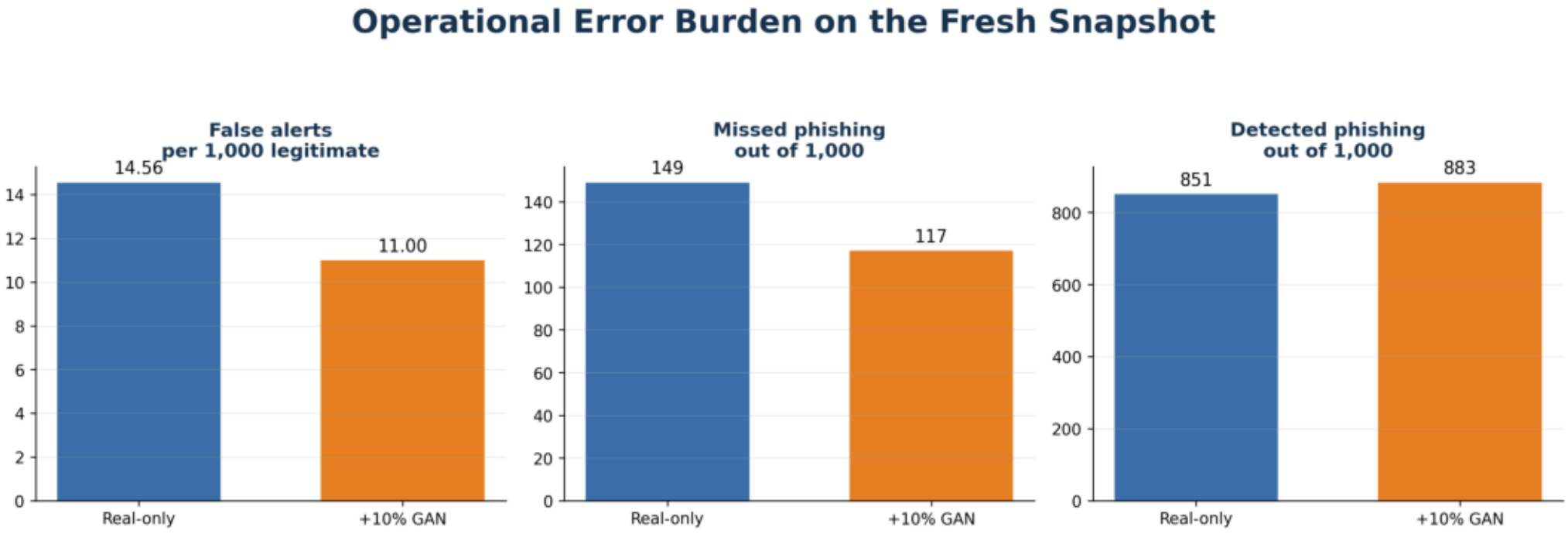
Paired decisions: GAN uniquely corrected 106 rows and uniquely worsened 42 rows; exact McNemar p = 1.446e-07.

This cohort was frozen before inference and was not used to alter thresholds or select checkpoints.

All Metrics


Fresh 10K Snapshot: Frozen Model Comparison
9,000 legitimate + 1,000 phishing URLs • thresholds unchanged
Accuracy
97.20%
97.84%
Precision
86.66%
89.92%
Recall
85.10%
88.30%
F1
85.87%
89.10%
ROC-AUC
97.03%
97.45%
PR-AUC
91.34%
93.51%
FPR
1.46%
1.10%
FNR
14.90%
11.70%
Real-only
+10% GAN

Operational Error Burden



Paired Uncertainty and Dataset Integrity

2,000 label-stratified paired bootstrap replicates

Accuracy	**+0.64%**	95% CI +0.41% to +0.88%
Precision	**+3.26%**	95% CI +1.61% to +5.04%
Recall	**+3.20%**	95% CI +1.90% to +4.60%
F1	**+3.23%**	95% CI +2.10% to +4.45%
ROC-AUC	**+0.41%**	95% CI +0.12% to +0.72%
PR-AUC	**+2.17%**	95% CI +1.58% to +2.81%
FPR	**-0.36%**	95% CI -0.59% to -0.16%
FNR	**-3.20%**	95% CI -4.60% to -1.90%

Frozen cohort

9,000 legitimate + 1,000 phishing • zero URL, hostname and 128-token overlap

SHA-256: cec7f6b43b3ebcbd12e345833489f603071459a551e997da749390e69d1a4ffe